Assignment 2

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Que 1

a construct the GNF from the given grammar :

$S \rightarrow XB \mid AA$

$A \rightarrow a \mid Sa$

$B \rightarrow b$

$X \rightarrow a$

⇒ consider

$S \rightarrow XB \mid AA$

put X is S

$\therefore S \rightarrow aB \mid AA$

put A in S

$\therefore S \rightarrow aB \mid (a \mid Sa)A$

$S \rightarrow aB \mid aA \mid SaA$

Remove left Recursion

$S' \Rightarrow aAS' \mid aA$

$S \rightarrow aBS' \mid aAS' \mid aB \mid aA$

Put S in A

$A \Rightarrow a \mid (aBS' \mid aAS' \mid aB \mid aA)a$

$A \rightarrow a \mid aBS'a \mid aAS'a \mid aBa \mid aAa$

because $X \rightarrow a$

$A \rightarrow a \mid aBS'X \mid aAS'X \mid aBX \mid aAX$

∴ Here is a GNF of given Grammar :

$S \Rightarrow aBS'X \mid aAS'X \mid aBX \mid aAX$
$A \rightarrow a \mid aBS'X \mid aAS'X \mid aBX \mid aAX$
$B \rightarrow b$
$X \rightarrow a$

Que 1
b construct the CNF. Consider the given grammar G as :

$S \rightarrow Aba \mid b$
$A \rightarrow bbA \mid B$
$B \rightarrow aAa$

⇒ consider $B \rightarrow aAa$
Let $B_1 \rightarrow a$

$B \rightarrow B_1AB_1$
Let $B_2 \rightarrow B_1A$

$B \rightarrow B_2B_1$

consider $A \rightarrow bbA \mid B$
Let $A_1 \rightarrow b$
& $B \rightarrow B_2B_1$

∴ $A \rightarrow A_1A_1A \mid B_2B_1$
Let $A_2 \rightarrow A_1A_1$

$A \rightarrow A_2A \mid B_2B_1$

consider $S \rightarrow Aba \mid b$
because $A_1 \rightarrow b$ & $B_1 \rightarrow a$

∴ $S \rightarrow AA_1B_1 \mid b$
Let $S_1 \rightarrow A_1B_1$

$S \rightarrow AS_1 \mid b$

Hence CNF production are

$S \rightarrow AS_1 \mid b$
$A \rightarrow A_2A \mid B_2B_1$
$B \rightarrow B_2B_1$

changes :-

- $B_1 \rightarrow a$
- $B_2 \rightarrow B_1A$

- $A_1 \rightarrow b$
- $A_2 \rightarrow A_1 A_1$
- $B \rightarrow B_2 B_1$
- $S_1 \rightarrow A_1 B_1$

Que 2

a construct the derivation tree from the following :

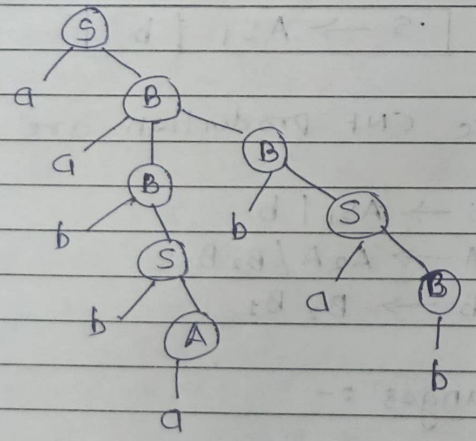
$S \rightarrow aB / bA$

$A \rightarrow a / as / bAA$

$B \rightarrow b / bs / aBB$

For the string · aabbabab

⇒ Derivation form in tree :



⇒ solved //

b construct a PDA to accept a language :

$L = \{ a^n . b^n \mid n > 0 \}$

⇒ so,

$n = 1, 2, 3 \ldots\ldots$

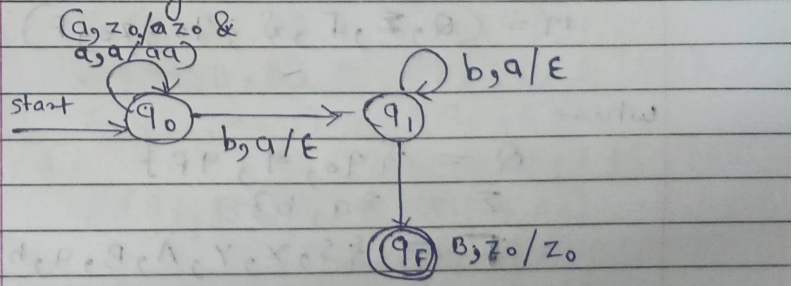
∴ string can be :-

ab, aabb, aaabbb ------

Transition Function :-

$\delta(q_0, a, z_0) = (q_0, az_0)$	ε
$\delta(q_0, a, a) = (q_0, aa)$	ε
$\delta(q_0, b, a) = (q_1, \epsilon)$	ε
$\delta(q_1, b, a) = (q_1, \epsilon)$	ε
$\delta(q_1, B, z_0) = (q_F, z_0)$ →	z_0

- State diagram :-



Therfore,

This PDA can accept any $\{a^n b^n \mid n>0\}$

like, ab, aabb, aaabbb.....

This the constucted PDA of

$L = \{a^n b^n \mid n>0\}$

Que 3 construct a PDA to accept following CFG.

$S \rightarrow XY$
$X \rightarrow AX / BX / a$
$Y \rightarrow YA / YB / a$
$A \rightarrow a$
$B \rightarrow b$

⇒ PDA :

$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$

where

$Q = \{q_0, q, q_F\}$
$\Sigma = \{a, b\}$
$\Gamma = \{S, X, Y, A, B, a, b, z_0\}$

Initial state = q_0
stack symbol = z_0
Final state = q_F

Transition function :

$\delta(q_0, \epsilon, z_0) = (q, Sz_0)$
$\delta(q, \epsilon, S) = (q_1, XY)$

For X :

$\delta(q_1, \epsilon, X) = (q_1, AX)$
$\delta(q_1, \epsilon, X) = (q_1, BX)$
$\delta(q_1, \epsilon, X) = (q_1, a)$

For Y :

$\delta(q_1, \epsilon, Y) = (q_1, YA)$
$\delta(q_1, \epsilon, Y) = (q_1, YB)$
$\delta(q_1, \epsilon, Y) = (q_1, a)$

For A, B :

$\delta(q_1, \epsilon, A) = (q_1, a)$
$\delta(q_1, \epsilon, B) = (q_1, b)$

$\delta(q_1, a, a) = (q_2, \epsilon)$
$\delta(q_2, b, b) = (q_2, \epsilon)$
$\delta(q_2, a, a) = (q_2, \epsilon)$

$\delta(q_2, B, z_0) = (q_F, z_0)$

∴ This PDA will accept the following CFG.

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Que 4

Q Explain Ackermanns function and calculate
A(1,1) , A(1,2) , A(2,1)

⇒ Ackermanns function are :

① A(0,y) = y+1
② A(x+1, 0) = A(x,1)
③ A(x+1, y+1) = A(x, A(x+1, y))

• A(1,1)

→ A(1,1) = A(0, A(0+1, 0))
= A(0, A(1,0)
= A(0, A(0,1))
= A(0, 2)
= 3

∴ A(1,1) = 3

• A(1,2)

→ A(1,2) = A(0, A(0+1, 1))
= A(0, A(1,1))
= A(0, 3)
= 4

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∴ A(1,2) = 4

• A(2,1)

→ A(2,1) = A(1, A(1+1, 0))
= A(1, A(2, 0))
= A(1, A(1, 1))
= A(1, 3)
= A(0, A(0+1, 2))
= A(0, A(1, 2))
= A(0, 4)
= 5

∴ A(2,1) = 5

∴ The following calculate Ackermanns functions are :

A(1,1) = 3
A(1,2) = 4
A(2,1) = 5

∴ This is the Explaination of Ackermanns function.

- Explaination :

- Ackermann's function is a recursive mathematical function that is defined using nested recursion.

Q4

b Explain the types of Turing machines Also explain the universal Turing machine.

⇒ Types of Turing machines.

1 Deterministic Turing machine (DTM):
- For each state and tape symbol there is exactly one possible transition.

2 Non-Deterministic Turing machine (NTM):
- For a state and tape symbol, multiple transitions may be possible.

3 Multi-Tape turing machine:
- uses more than one tape, with each tape having its own read/write head.

4 multi-Track Turing machine:
- A single tape is divided into multiple tracks, allowing multiple symbols to be stored at each tape position.

5 universal Turing machine (UTM):
- A Turing machine that can simulate the operation of any other Turing machine.

Universal turing machine:

- A UTM takes the description of another Turing machine and its input as input.

$$U(\langle M \rangle, w) = M(w)$$

where,
$\langle M \rangle$ is encoded description
w is input string.

Thus, a UTM acts as a general purpose Turing machine capable of simulating other Turing machines.

Ques

Q Construct a Turing machine for the Language :

$$L = \{ 0^n 1^n \mid n \geq 1 \}$$

⇒ Approach :

The turing machine matches each '0' with one '1'.

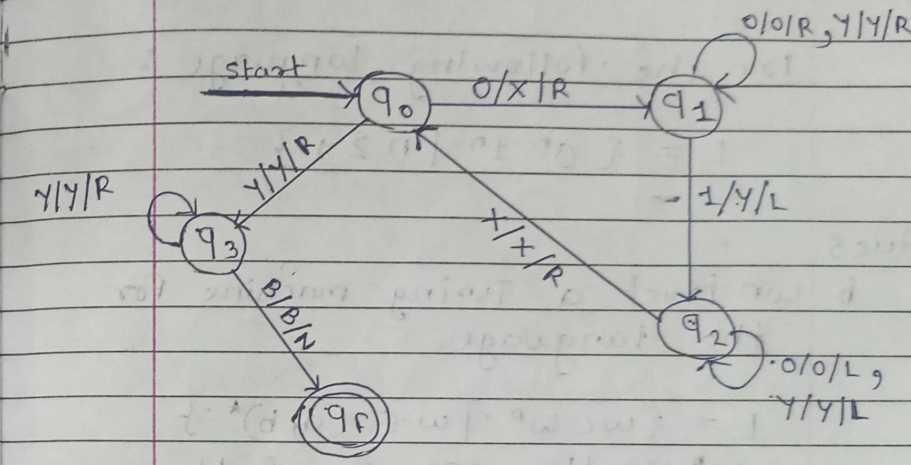
- Replace the first unmarked '0' by X.
- Move right and find first unmarked 1.
- Replace it by Y.
- Return to the left and Repeat.
- If all '0's and '1's are matched correctly, accept.
- Otherwise, reject.

States :

$$Q = \{ q_0, q_1, q_2, q_3, q_f \}$$

where, q_0 = initial state
q_f = final state
$q_{1,2,3}$ = other states

0	0	0	0	1	1	1	1	B
X	X	X	X	Y	Y	Y	Y	



★ Transition table :

present state	Input tape symbol				
	0	X	1	Y	B
q_0	q_1, X/R			q_3, Y/R	
q_1	q_1, 0/R		q_3, Y/L	q_1, Y/R	
q_2	q_2, 0/L	q_0, X/R		q_2, Y/L	
q_3				q_3, Y/R	q_f, B/N
q_f	—	—	—	—	—

∴ This are The turning machine with it transition state table.

For the following language :

$L = \{ 0^n 1^n \mid n \geq 1 \}$

Que 5

b construct a Turing machine for the language

$L = \{ w c w^R \mid w \in (a,b)^* \}$

w^R is the reserse of w

⇒ Given,

$(a,b)^*$

w^R are reverse of w

∴ a c a

b c b

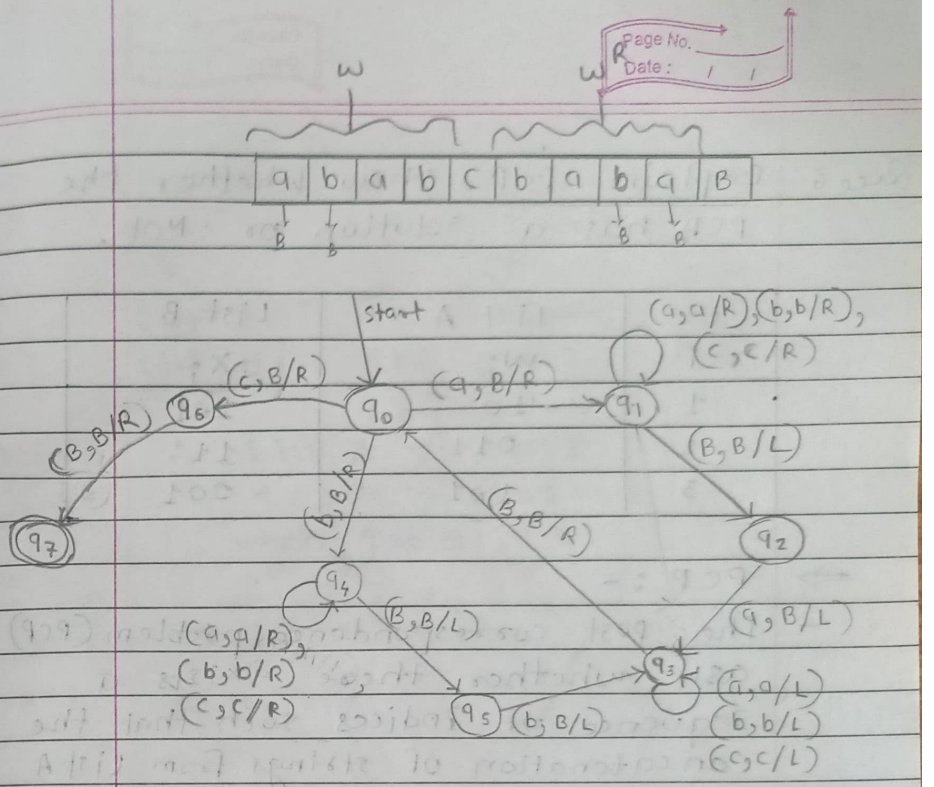
a b c b a

a b a b c b a b a

w c w^R ⇒ $w c w^R$

&

$w \in (a,b)^*$



★ Transition table

present state	Input tape symbol			
	a	b	c	B
q_0	$q_1, B/R$	$q_4, B/R$	$q_6, B/R$	
q_1	$q_1, a/R$	$q_1, b/R$	$q_1, c/R$	$q_2, B/L$
q_2	$q_3, B/L$			
q_3	$q_3, a/L$	$q_3, b/L$	$q_3, c/L$	$q_0, B/R$
q_4	$q_4, a/R$	$q_4, b/R$	$q_4, c/R$	$q_5, B/L$
q_5		$q_3, B/L$		
q_6				$q_7, B/R$
q_7	—	—	—	—

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Que 6 Explain PCP. Show whether the PCP has a solution or not.

	List A	List B
i	w_i	x_i
1	10	101
2	011	11
3	101	001

⇒ PCP :-

- The post correspondence problem (PCP) ask whether there exists a sequence of indices such that the concatenation of strings from List A is equal to the concatenation of strings from List B.

(1) Consider i = 1

i :	1	1
List A :	10	10
List B :	101	101

∴ List A ≠ List B
1010 ≠ 101101
when i = 1

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consider i = 1

i :	1	3
List A :	10	101
List B :	101	001

∴ List A ≠ List B
10101 ≠ 101001
when i = 3

(2) consider i = 2

i :	2
List A :	011
List B :	11

∴ List A ≠ List B
011 ≠ 11
when i = 2

(3) consider i = 3

i :	3
List A :	101
List B :	001

$\therefore$ List A $\neq$ List B

$101 \neq 001$

when $i = 3$

$\therefore$ The given PCP has No solution.

